Rethinking case attraction on Ancient Greek infinitive clauses

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1 Introduction

The variation of constructions with (1-a) and without (1-b) case attraction on arguments of infinitive clauses in Ancient Greek remains largely unexplained. The more philological driven efforts have only tentatively drawn contexts in which a certain option is more likely to occur, a handful of times assuming that such correlation between context and resolution was causal, often due to an abstract concept of *emphasis*, without any empirical evidence. Other works have simply asserted its existence. I would enroll my own works (Geraldes 2020, 2021, 2025) in the first group, as I mainly argued for a more precise system of correlations, providing quantitative evidence for such.

(1) a. ἀφῆκε μοι ἐλθόντι πρὸς ὑμᾶς λέγειν
allowed.3SG PRON.1SG.DAT going.DAT.SG in-front-of-you say.INF
τἀληθῆ.
the-truth.ACC.
He allowed me to go and speak the truth in front of you. (Xen. Hell. 6.1.13)

b. συμβουλεύει τῷ Ξενοφῶντι ἐλθόντα εἰς Δελφοὺς ἀνακοινῶσαι
advice.3SG X.DAT.SG going.ACC.SG to-Delphi ask.INF
τῷ θεῷ περὶ τῆς πορείας.
the-god.DAT.SG about-the-travel
He advices Xenophon to go to Delphi and ask the god about the travel. (Xen. Anab. 3.1.5)

A third group, that of structurally driven works, most often from the generative or minimalist framework, has proposed structures that would allow the attraction to occur, in just a handful of them proposing a specific structure that would allow for the non-attraction. Most recently, Sevdali (2013a,b) (drawing part of their analysis from the works of Tantalou 2003 and Spyropoulos 2005) argued that case attraction,

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or *case transmission* as they call, is as a byproduct of the syntactic marking of $[Person]$ in a otherwise *person*-deficient clause, i.e. infinitive clauses. In other words, they assume the existence of a deficient Person head projected over the infinitive verb with a caseless PRO, which therefore would probe for case until if finds the oblique object of the matrix clause, and finally provides case for its predicate when it probes for case. The existence of (1-b) is explained by an optional accusative *pro* acting as subject for the infinitival clause which the author argues is available in Ancient Greek when it refers to an element "prominent in the discourse but not present in the clause that immediately contains the infinitive" (Sevdali 2013b, pp. 42–3). Similarly to the "emphasis" explanation, however, the "discourse prominence" does not comes with empirical evidence or clear definition.

2 A stoachastic approach

Using data from literary sources of Classical Greek, including oratory speeches, drama, historiography and philosophical dialogues from Attic and Jonic sources, I provide a data driven quantitative analysis of the contexts in which case attraction is a possible agreement / concord resolution. The addition of Jonic sources is due to the fact that it has been assumed that case attraction is more common if not the rule in the Attic dialect (e.g. Buttmann 1826, *passim.* and Cooper and Krüger 1997, *ad loc.*). The data has been collected and annotated in a combination of manual and computational methods using the Diorisis Ancient Greek Corpus (Vatri and McGillivray 2018).

A quantitative analysis of case attraction requires caution in the methodological approach, as semantic and pragmatic features are often latent, i.e. not explicitly present in the morpho-phonological level, and the use of proxies such as word classes, constituent distance and word order may hinder the quality of the results and the causal inference built upon them. As such, our analysis will rely on the causality analysis (e.g. Pearl 2009) and Bayesian modeling (e.g. McElreath 2020), which I argue could enhance the results of data driven linguistic research by including at the quantitative analysis the qualitative knowledge built on the Ancient Greek and general linguistics. The analysis must be thus twofold. Firstly, it assess how the linguistic and extralinguistic factors could reasonably be causally linked with case attraction, so as to inform the statistical modeling and tell what effects are possible to estimate from the data. Later, a general linear model is build as to adequately estimate the direct effects of semantic and pragmatic factors on case attraction and the interactions between linguistic and extralinguistic variables.

3 A causal model for case attraction

Case attraction consists in a covariance of a specific morphological feature (case) between a controller (the oblique argument of a certain matrix verb) and a target (the predicative or attributive expression modifying the logical subject of the infinitive verb). Thus, it falls into the general concept of *agreement*, in Corbett's framework (2006, pp. 4–7).

By *agreement*, we mean the general class of phenomena by which morphological features covary across a pair of tokens with a particular direction, i.e.: in which there are a controller and a target. Even there are indeed, as Norris (2014, 2017) and Polinsky (2016) argue, different mechanisms behind subject-verb agreement, DP-internal concord, and predicate case concord – being case attraction an instance of the last one –, our use of Corbett's framework will remain unaltered, as the differences observed between DP-internal concord and subject-verb agreement in the authors defending the contrasts are irrelevant for our analysis.

By assuming that case attraction (A) is a sort of *non-canonical agreement/concord*, it must be expected that factors known cross-linguistically to trigger these *phenomena* also affect A. Drawing a list of conditioning factors for *non-canonical agreement* from Corbett (2006, pp. 176–205), we should thus expect that the following factors would change the likelihood of A. 1. thematic role (Θ) and animacy (An); 2. grammatical relation (GR) and case (K); and 3. communicative function (CF), and precedence ($Prec$).

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A Criteria for Canonical Agreement

The following list summarizes the criteria for canonical agreement as discussed in section 1.4 of Corbett (2006, pp. 8ff.). The underlines indicate what position in the hierarchy case attraction arguably occupies.

- Controllers:
  - C-1: controller present > controller absent
  - C-2: controller has overt expression of agreement features > controller has covert expression of agreement features
    - “A canonical controller marks at least as many distinctions as the target”
  - C-3: consistent controller > hybrid controller
  - C-4: controller’s part of speech is irrelevant > is relevant (given the domain)
- Targets:
  - C-5: inflectional marking (affix) > clitic > free word
  - C-6: obligatory > optional
  - C-7: regular > suppletive
  - C-8: alliterative > opaque
  - C-9: productive marking of agreement > sporadic marking
  - C-10: target always agree > target agrees only when controller is absent
  - C-11: target agrees with a single controller > agrees with more than one controller
  - C-12: target has no choice of controller > target has choice of controller (is ‘trigger-happy’)
  - C-13: target’s part of speech is irrelevant > is relevant (given the domain)
- Domains:
  - C-14: asymmetric > symmetric
  - C-15: local domain > non-local domain
  - C-16: domain is one of a set > single domain

- Features:
  - C-17: feature is lexical > non-lexical
  - C-18: features have matching values > non-matching values
  - C-19: no choice of feature value > choice of value
- Conditions:
  - C-20: no conditions > conditions